

Instructions

- Answer all eight questions. Show sufficient working for method marks.
- A calculator may be used unless a question states otherwise.
- Write clearly in the spaces provided. Additional pages may be attached.

SCORE SAVED TO COURSE PROGRESS

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1

Explain how the structure of an alveolus is adapted for gas exchange.

[5 marks]

Method sound; final simplification omitted.

5/5

2

Outline the process of photosynthesis and state the word equation.

[5 marks]

Check the condition before applying the rule.

5/5

AGENT ANNOTATED

3

Describe how a reflex arc produces a rapid response, naming the neurones involved.

[6 marks]

Correct and clearly justified.

5/6

4

Explain how natural selection can lead to antibiotic resistance in bacteria.

[6 marks]

Method sound; final simplification omitted.

5/6

5

Describe an experiment to investigate the effect of light intensity on the rate of photosynthesis.

[6 marks]

Check the condition before applying the rule.

6/6

AGENT ANNOTATED

6

Explain the difference between mitosis and meiosis and state where each occurs.

[5 marks]

Correct and clearly justified.

4/5

7

Use a Punnett square to predict the offspring ratio from two heterozygous parents.

[4 marks]

Method sound; final simplification omitted.

4/4

8

Describe how the body maintains a constant internal temperature.

[5 marks]

Check the condition before applying the rule.

4/5